

DAY 16 – Q16 – Web API and EF Core

Q1. Which return type allows returning both strongly typed objects and HTTP status codes from a Web API action?

- A. object
- B. IActionResult
- C. ActionResult
- D. JsonResult

Answer: C

Q2. Which method actually commits pending changes to the SQL Server database?

- A. Update()
- B. Commit()
- C. SaveChanges()
- D. Execute()

Answer: C

Q3. Which EF Core method retrieves an entity by its primary key?

- A. First()
- B. Find()
- C. Single()
- D. Last()

Answer: B

Q4. Which HTTP status code should be returned after successfully creating a new resource?

- A. 200
- B. 201
- C. 204
- D. 302

Answer: B

Q5. Which interface is typically injected into a controller when using Repository Pattern?

- A. DbContext
- B. Repository class directly
- C. Repository Interface
- D. SqlConnection

Answer: C

Q6. Find the Error in code below

```
public IActionResult Get(int id)
{
    var emp = _context.Employees.Find(id);
    return Ok;
}
```

- A. Find() cannot be used
- B. Ok should be Ok()
- C. IActionResult is invalid
- D. id should be string

Answer: B

Q7. DTOs are primarily used to

- A. Improve SQL Server performance
- B. Hide internal entity structure
- C. Replace Controllers
- D. Replace DbContext

Answer: B

Q8. Fill in the Code

```
services.AddDbContext<AppDbContext>(
options => options._____ (connectionString));
```

- A. UseMySQL
- B. UseOracle
- C. UseSqlServer
- D. UseSQLiteMemory

Answer: C

Q9. (Find the Error)

```
[HttpGet]
public Employee GetEmployee(int id)
{
    return NotFound();
}
```

- A. HttpGet is invalid

- B. Return type should allow NotFound()
- C. int id cannot be used
- D. Employee class is invalid

Answer: B

Q10. Fill in the Code

```
public class EmployeeDto
{
    public int Id { get; set; }

    [_____]
    public string Name { get; set; }
}
```

- A. Validate
- B. Required
- C. Mandatory
- D. NotNull

Answer: B

Q11. Find the Error

```
public async Task<IActionResult> Save(Employee emp)
{
    _context.Employees.Add(emp);
    return Ok();
}
```

- A. Add() is incorrect
- B. SaveChangesAsync() is missing
- C. IActionResult is invalid
- D. Employee should be DTO

Answer: B

Q12. Fill in the Code

```
var employee = await _context.Employees
    ._____(e => e.Department)
    .ToListAsync();
```

- A. Attach
- B. Join
- C. Include
- D. Load

Answer: C

Q13. Which method is used to check whether an entity exists in EF Core?

- A. Exists()
- B. Contains()
- C. Any()
- D. Find()

Answer: C

Q14. Which Data Annotation specifies the primary key of an entity?

- A. [Primary]
- B. [Identity]
- C. [Key]
- D. [Id]

Answer: C

Q15. Find the Error

```
public async Task<IActionResult> GetEmployees()
{
    var employees = await _context.Employees.ToListAsync();
    return employees;
}
```

- A. ToListAsync() is invalid
- B. employees should be Employee[]
- C. Should return Ok(employees)
- D. IActionResult cannot be async

Answer: C

Q16. Fill in the Code

```
public async Task<Employee?> GetById(int id)
{
    return await _context.Employees._____(e => e.Id == id);
}
```

```
}
```

- A. FirstOrDefaultAsync
- B. ToListAsync
- C. FindAllAsync
- D. SingleAsyncList

Answer: A

Q17. Fill in the Code

```
if (!ModelState._____)
{
    return BadRequest(ModelState);
}
```

- A. Success
- B. Valid
- C. IsValid
- D. Validate

Answer: C

Q18. Which Data Annotation validates the minimum and maximum length of a string?

- A. [Range]
- B. [StringLength]
- C. [Length]
- D. [Size]

Answer: B

Q19. (Fill in the Code)

```
[HttpGet("{id}")]
public async Task<ActionResult<EmployeeDto>> Get(int id)
{
    var employee = await _repository._____(id);
}
```

- A. GetByIdAsync
- B. FindAll
- C. Select
- D. Load

Answer: A

Q20. Which of the following is the best practice for asynchronous repository methods?

- A. Return void
- B. Return Task
- C. Return dynamic
- D. Return object

Answer: B